

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CE/IT)

March 2018

IT402/IT402.01: Wireless Communication & Mobile Computing

Date: 30.03.2018, Friday

Time: 10:00 am To 01:00 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Write the answer of following Questions. [11]

- (a) Compare Wireless & Wired Communication. [2]
- (b) What is 802.11? [1]
- (c) What is Active v/s Passive hotspot? [1]
- (d) What is the need of CSMA/CA protocol? [1]
- (e) In Mobile IP, Co-located Care of Address, _____ acts as a tunnel end point. [1]
(a) Foreign Agent (b) Home Agent (c) Mobile Agent (d) Correspondent Agent.
- (f) What is use of TMSI and MSRN numbers? [1]
- (g) What is Inter Frame Spacing with respect to Wireless LAN? [2]
- (h) Briefly explain following terms in the context of Wireless Lan. [2]
1) Open Access Authentication 2) Shared Key Authentication

Q - 2 Write the answer of following Questions. [12]

- (a) Explain Direct sequence spread spectrum. [5]
- (b) Explain WATM reference model. [7]

OR

- (b) What are the advantages and disadvantages of Wireless LAN? [7]

Q - 3 Write the answer of following Questions. (Attempt any TWO) [12]

- (a) Explain Time Division Multiple Access (TDMA).
- (b) Explain FDMA.
- (c) Explain Wireless Application Protocol.

SECTION – II

Q - 4 Write the answer of following Questions. [11]

- (a) List the disadvantages of Snooped TCP. [2]
- (b) Explain 802.11 Physical layer. [1]
- (c) Differentiate Infrared and Bluetooth. [1]
- (d) _____ number of active addresses are possible in Bluetooth? [2]
(a) 7 (b) 8 (c) 15 (d) 16
- (e) Explain typed of Wireless LAN. [2]
- (f) How the movement of mobile node affects routing in wireless network? [3]

Q - 5 Write the answer of following Questions. [12]

- (a) Explain Modulation and Multiplexing with respect to application. [5]
- (b) Explain GSM architecture. [7]

OR

- (b) Explain GPRS in detail. [7]

Q - 6 Write the answer of following Questions. (Attempt any TWO) [12]

- (a) Explain Elements of DECT Architecture.
- (b) Explain HiperLan.
- (c) Explain Bluetooth.